Day 3 – Searching Algorithms (Java DSA)

1️⃣ Searching kya hota hai?

Searching ka matlab hai array ya list me kisi element ko find karna.

Example Array

10 20 30 40 50

Agar hume 30 search karna hai, to algorithm batayega ki 30 kis index par hai.

2️⃣ Linear Search

Linear search me array ke har element ko ek-ek karke check kiya jata hai.

Example

Array = 10 20 30 40 50
Key = 30

Steps

10 check karo

20 check karo

30 check karo

Element mil gaya

Java Code – Linear Search

```
public class LinearSearch {

    public static void main(String[] args) {

        int[] arr = {10,20,30,40,50};
        int key = 30;

        for(int i = 0; i < arr.length; i++){

            if(arr[i] == key){
                System.out.println("Element found at index " + i);
            }

        }

    }

}
```

Output

Element found at index 2

Linear Search Complexity

Case	Complexity
Best Case	O(1)
Average Case	O(n)
Worst Case	O(n)

Example

Agar element first position par mil gaya → Best Case

Agar last position par hai → Worst Case

3️⃣ Binary Search

Binary Search sorted array par hi kaam karta hai.

Example

2 5 8 10 15 20

Binary search me array ko repeatedly half kiya jata hai.

Binary Search Steps

Example

Array = 2 5 8 10 15 20
Key = 10

Steps

Middle element find karo

Agar key == mid → element mil gaya

Agar key < mid → left side search

Agar key > mid → right side search

Java Code – Binary Search

```
public class BinarySearch {

    public static void main(String[] args) {

        int[] arr = {2,5,8,10,15,20};

        int key = 10;

        int left = 0;
        int right = arr.length - 1;

        while(left <= right){

            int mid = (left + right) / 2;

            if(arr[mid] == key){

                System.out.println("Element found at index " + mid);
                break;

            }

            else if(arr[mid] < key){

                left = mid + 1;

            }

            else{

                right = mid - 1;

            }

        }

    }

}
```

Output

Element found at index 3

Binary Search Complexity

Case	**Complexity**
Best Case	O(1)
Average Case	O(log n)
Worst Case	O(log n)

Example

Agar array size 16 hai

Search steps

16 → 8 → 4 → 2 → 1

Isliye complexity:

O(log n)

4️⃣ Linear vs Binary Search

Feature	Linear Search	Binary Search
Array sorted hona zaruri	❌ No	✅ Yes
Speed	Slow	Fast
Complexity	O(n)	O(log n)

5️⃣ Day 3 Practice Problems

Problem 1

Linear search implement karo.

Problem 2

Binary search implement karo.

Problem 3

Array me largest element find karo.

Problem 4

Array me element exist karta hai ya nahi check karo.